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4.2 Bonding, Structure and the Properties of Matter

AQA GCSE Chemistry (8462) — Paper 1

4.2.1 Chemical Bonds: Ionic, Covalent and Metallic

Chemists use theories of structure and bonding to explain the physical and chemical properties of materials. Analysing structures shows that atoms can be arranged in a variety of ways, some of which are molecular while others are giant structures. Understanding bonding is the key to explaining why salt has a high melting point, why oxygen is a gas, and why copper conducts electricity.

4.2.1.1 The Three Types of Strong Chemical Bond

Bond type	Occurs between	Particles involved	What happens to electrons
Ionic	A **metal** and a **non-metal**	Oppositely charged ions	Electrons are **transferred** from the metal atoms to the non-metal atoms.
Covalent	**Non-metals** only (in compounds or elements)	Atoms that share pairs of electrons	Pairs of electrons are **shared** between atoms.
Metallic	Atoms in **metals and alloys**	Positive metal ions and delocalised electrons	Outer electrons are **delocalised** (free to move through the whole structure).

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Exam Tip: You should be able to explain chemical bonding in terms of electrostatic forces and the transfer or sharing of electrons. A quick way to decide the bond type from a formula: metal + non-metal (e.g. NaCl, MgO, CaF_2) → ionic; only non-metals (e.g. H_2O, CO_2, CH_4, Cl_2) → covalent; pure metal or alloy (e.g. Cu, brass) → metallic.

4.2.1.2 Ionic Bonding

When a **metal** atom reacts with a **non-metal** atom, **electrons in the outer shell of the metal atom are transferred** to the non-metal atom. Metal atoms lose electrons to become **positively charged ions**; non-metal atoms gain electrons to become **negatively charged ions**. The ions produced have the **electronic structure of a noble gas** (Group 0) — a full outer shell.

The charge on an ion formed by an element in Groups 1, 2, 6 and 7 relates directly to its group number:

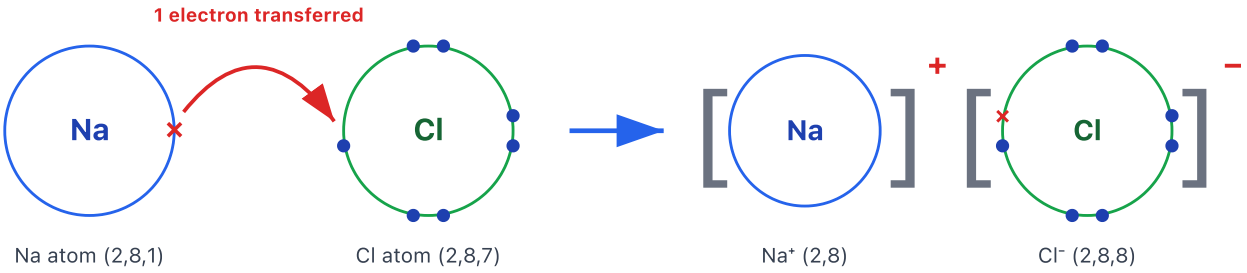
Group	Outer electrons	Electrons lost/gained	Ion charge	Examples
1	1	Loses 1	1+	Li^+, Na^+, K^+
2	2	Loses 2	2+	Mg^{2+}, Ca^{2+}
3	3	Loses 3	3+	Al^{3+}
6	6	Gains 2	2–	O^{2-} (oxide), S^{2-} (sulfide)
7	7	Gains 1	1–	F^-, Cl^- (chloride), Br^-, I^-

Dot and Cross Diagrams for Ionic Compounds

The electron transfer during the formation of an ionic compound can be represented by a **dot and cross diagram**. Electrons from one atom are drawn as dots and from the other as crosses, so you can see where each electron came from. The final ions are drawn in square brackets with the charge at the top right. Often only the **outer shell** is drawn.

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Formation of sodium chloride (outer shells only)



Sodium loses its outer electron (×) to chlorine. Both ions now have full outer shells (noble gas structures). Na^+ has the electronic structure of neon; Cl^- has the electronic structure of argon.

Na^+'s now-empty third shell is not drawn — its outer shell is the full second shell (8 electrons, not shown).

Working Out Formulae of Ionic Compounds

An ionic compound has **no overall charge**, so the total positive charge must equal the total negative charge. Combine ions in the ratio that makes the charges cancel.

Compound	Ions	Ratio for zero charge	Formula	Electron transfer
Sodium chloride	Na^+, Cl^-	1 : 1	NaCl	1 electron from Na to Cl
Magnesium oxide	Mg^{2+}, O^{2-}	1 : 1	MgO	2 electrons from Mg to O
Calcium chloride	Ca^{2+}, Cl^-	1 : 2	$CaCl_2$	1 electron from Ca to each of 2 Cl atoms
Sodium oxide	Na^+, O^{2-}	2 : 1	Na_2O	1 electron from each of 2 Na atoms to O
Aluminium oxide	Al^{3+}, O^{2-}	2 : 3	Al_2O_3	2 Al lose 6 electrons; 3 O gain 6 electrons

Some ions contain more than one element (**compound ions**). You should know: hydroxide **OH^-**, nitrate **NO_3^-**, sulfate **SO_4^{2-}**, carbonate **CO_3^{2-}** and ammonium **NH_4^+**. Use brackets when more than

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one compound ion is needed: calcium hydroxide is $Ca(OH)_2$; ammonium sulfate is $(NH_4)_2SO_4$.

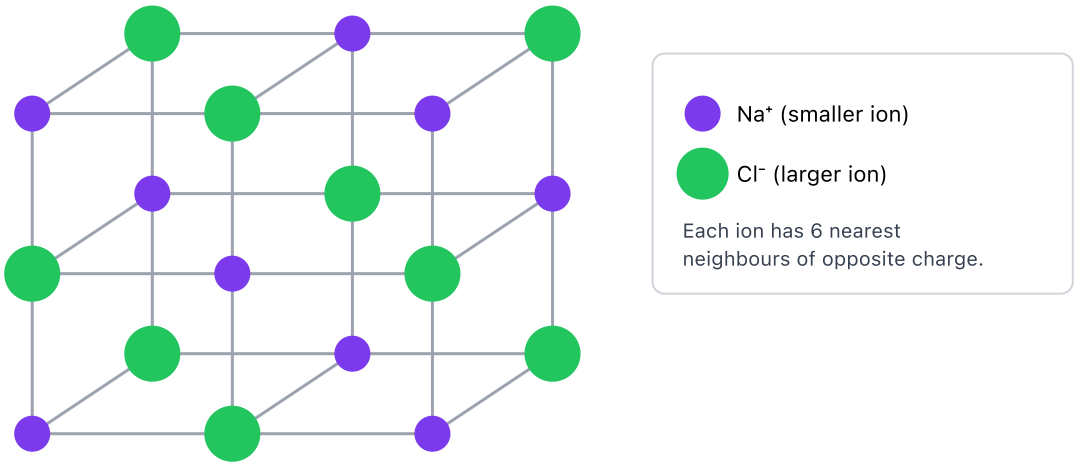
> **Exam Tip:** In ionic dot and cross diagrams, examiners look for: (1) the **correct number of electrons** in the outer shell of each ion (usually 8); (2) **square brackets** around each ion; (3) the **correct charge** on each ion; (4) dots and crosses making clear which electrons were transferred. For $CaCl_2$ or Na_2O you must draw **two** of the ion that appears twice (or write "2" in front of it).

4.2.1.3 Ionic Compounds

> An ionic compound is a **giant structure of ions**. Ionic compounds are held together by **strong electrostatic forces of attraction between oppositely charged ions**. These forces act in **all directions** in the lattice, and this is called **ionic bonding**.

The structure of sodium chloride is a regular, repeating cubic lattice. Each Na^+ ion is surrounded by six Cl^- ions, and each Cl^- ion is surrounded by six Na^+ ions. The lattice extends in three dimensions and contains a huge number of ions — there is no such thing as a single "molecule" of sodium chloride. The formula NaCl simply tells us the **ratio** of ions (1 : 1).

Sodium chloride: a giant ionic lattice (ball-and-stick model)



The lattice continues in all directions. The "sticks" are not real bonds — they show positions of ions.

Models of Ionic Structures and their Limitations

Model	Advantages	Limitations
Dot and cross diagram	Shows how ions form and where electrons come from; shows charges.	Does not show the 3D arrangement of ions or relative sizes; implies electrons from different atoms are different (they are identical).
Ball-and-stick (3D)	Shows the regular 3D arrangement of ions in the lattice.	Large gaps between ions (in reality ions are touching); "sticks" suggest bonds in fixed directions, but ionic forces act in all directions; only shows a small part of the lattice.
Space-filling (close-packed)	Shows relative sizes and that ions are closely packed.	Hard to see the arrangement inside; only shows outer layer.
2D diagram	Simple; easy to draw; shows alternating ions.	Does not show the third dimension; does not show how ions are held together.

Deducing the Empirical Formula from a Model

You should be able to work out the empirical formula of an ionic compound from a diagram of its structure: count the ions of each type (or use their charges) and simplify to the lowest whole-number ratio. For example, a 2D diagram showing 8 Mg^{2+} ions and 8 O^{2-} ions gives MgO; a diagram showing 6 K^{+} and 3 O^{2-} gives K_2O.

4.2.1.4 Covalent Bonding

> When atoms share **pairs of electrons**, they form **covalent bonds**. These bonds between atoms are **strong**. Covalent bonding occurs in most **non-metallic elements** and in **compounds of non-metals**. Each shared pair of electrons is one covalent bond; the shared electrons are attracted to the nuclei of both atoms.

Covalently bonded substances may consist of:

- **small molecules** — e.g. H_2, Cl_2, O_2, N_2, HCl, H_2O, NH_3, CH_4;
- **large molecules** — such as polymers (e.g. poly(ethene));
- **giant covalent structures** (macromolecules) — such as diamond, graphite and silicon dioxide.

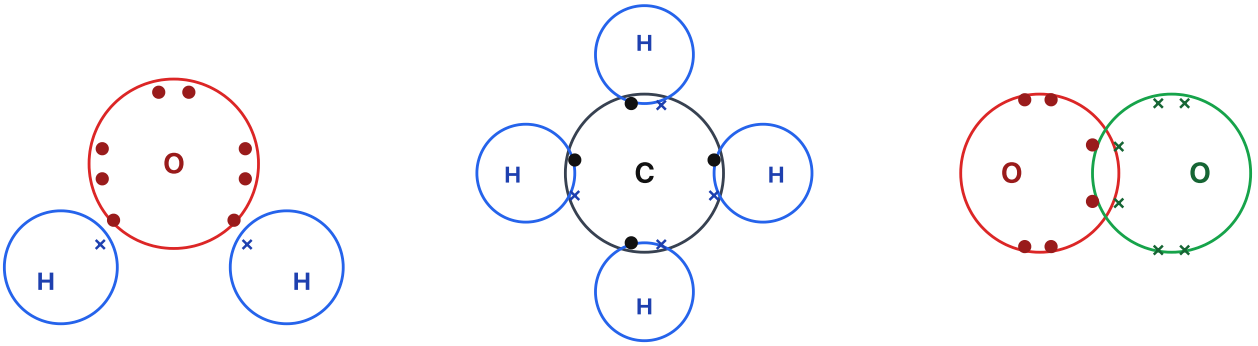
Dot and Cross Diagrams for Small Molecules

You must be able to draw dot and cross diagrams for the molecules H_2, Cl_2, O_2, N_2, HCl, H_2O, NH_3 and CH_4. The shared electrons are drawn in the overlap of the two atoms' outer shells.

Molecule	Displayed formula	Bonds	Lone pairs on central atom
Hydrogen, H_2	H–H	1 single bond	—
Chlorine, Cl_2	Cl–Cl	1 single bond	3 on each Cl
Oxygen, O_2	O=O	1 **double** bond (2 shared pairs)	2 on each O
Nitrogen, N_2	N≡N	1 **triple** bond (3 shared pairs)	1 on each N
Hydrogen chloride, HCl	H–Cl	1 single bond	3 on Cl
Water, H_2O	H–O–H	2 single bonds	2 on O
Ammonia, NH_3	N bonded to 3 H	3 single bonds	1 on N
Methane, CH_4	C bonded to 4 H	4 single bonds	0

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Dot and cross diagrams for covalent molecules (outer shells)



Water, H_2O
2 bonding pairs, 2 lone pairs

Methane, CH_4
4 bonding pairs, no lone pairs

Oxygen, O_2 (O=O)
Double bond: 2 shared pairs

Representing Covalent Substances and their Limitations

Representation	What it shows	Limitations
Dot and cross	Which atoms the bonding electrons came from; lone pairs.	Does not show the shape of the molecule or relative atom sizes.
Displayed formula (lines for bonds, e.g. H–O–H)	Which atoms are bonded and how many bonds.	Does not show the 3D shape or which electrons came from which atom; no lone pairs shown.
Ball-and-stick (3D)	The 3D shape and bond angles.	Atoms are not really separated by long gaps; bonds are not physical sticks; relative sizes not accurate.
Space-filling	Relative sizes and overall shape.	Hard to see bonds and bond angles.

You should be able to deduce the **molecular formula** of a substance from a given model or diagram in these forms, by counting the atoms of each element. For a giant covalent structure, the formula is an empirical (ratio) formula, e.g. SiO_2.

> **Exam Tip:** In covalent dot and cross diagrams, each atom must end up with a full outer shell (8 electrons, or 2 for hydrogen). Shared pairs go in the overlap. A common error is forgetting

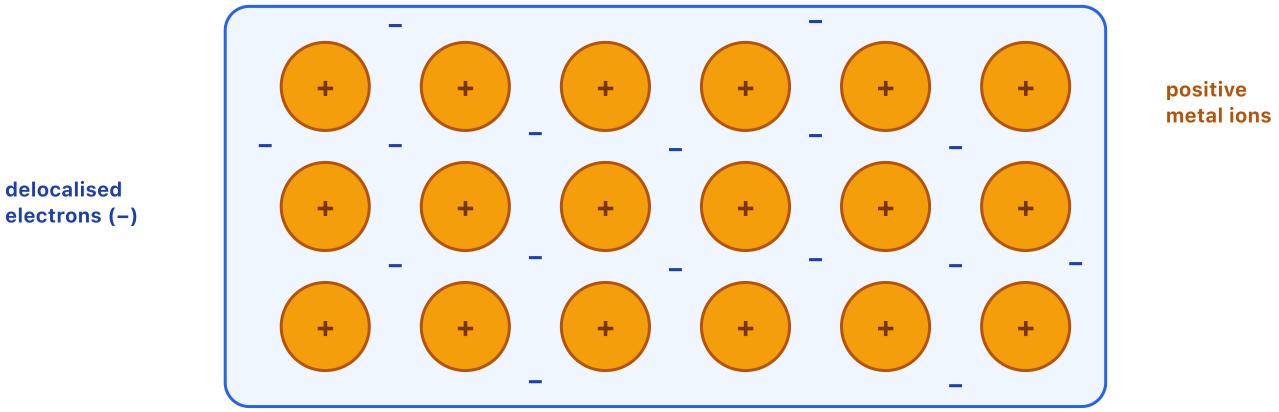
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lone pairs — e.g. drawing NH_3 with only 6 electrons around nitrogen. Count every shell after drawing. For N_2, there are 3 shared pairs (6 shared electrons) plus 1 lone pair on each N.

4.2.1.5 Metallic Bonding

Metals consist of **giant structures of atoms arranged in a regular pattern**. The electrons in the outer shell of metal atoms are **delocalised** and so are free to move through the whole structure. The **sharing of delocalised electrons** gives rise to **strong metallic bonds** — the strong electrostatic attraction between the positive metal ions and the "sea" of delocalised electrons.

Metallic bonding: positive ions in a sea of delocalised electrons



Ions are arranged in regular layers. Electrons move freely throughout the whole structure.

4.2.2 How Bonding and Structure are Related to the Properties of Substances

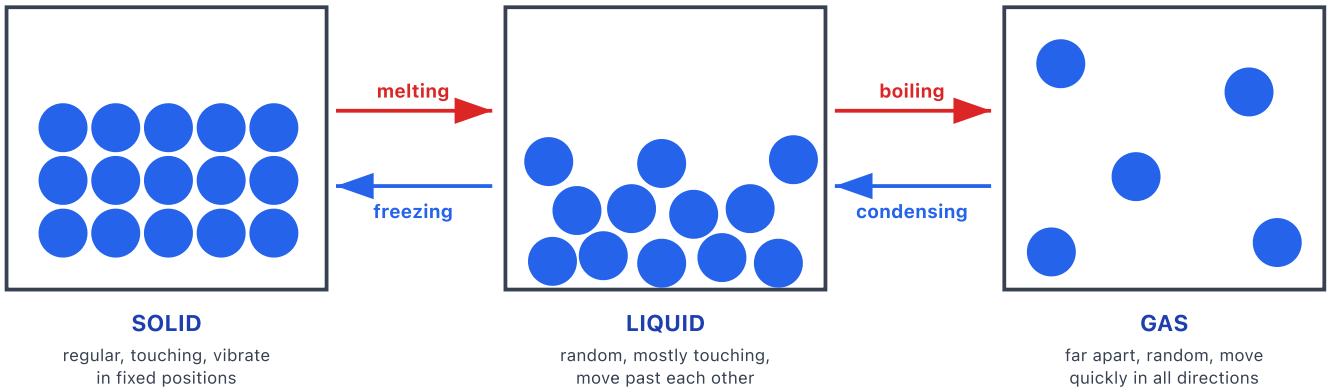
4.2.2.1 The Three States of Matter

The three states of matter are **solid**, **liquid** and **gas**. **Melting** and **freezing** take place at the **melting point**; **boiling** and **condensing** take place at the **boiling point**. The three states can

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be represented by a simple model in which particles are represented by small solid spheres. Particle theory can help to explain melting, boiling, freezing and condensing.

The particle model of solids, liquids and gases



The **amount of energy needed to change state** from solid to liquid and from liquid to gas depends on the **strength of the forces between the particles** of the substance. The nature of the particles involved depends on the type of bonding and the structure of the substance. The **stronger the forces** between the particles, the **higher the melting point and boiling point** of the substance.

Predicting States from Data

To predict the state of a substance at a given temperature:

- Temperature **below** the melting point → **solid**.
- Temperature **between** the melting point and boiling point → **liquid**.
- Temperature **above** the boiling point → **gas**.

Worked Example 1 — Predicting the State

```
Question: Substance X has a melting point of −7 °C and a boiling
          point of 59 °C. What is its state at 25 °C and at 80 °C?

At 25 °C:  −7 < 25 < 59   → between m.p. and b.p. → LIQUID
At 80 °C:  80 > 59         → above b.p.             → GAS

(Substance X is bromine.)
```

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Limitations of the Simple Particle Model (Higher)

The particle model represents particles as small, solid, inelastic spheres. This is useful but has limitations:

- There are **no forces** shown between the spheres, so the model cannot explain why different substances have different melting and boiling points.
- All particles are represented as **spheres**, but real particles may be atoms, ions or molecules of many different shapes.
- The particles are shown as **solid** — real atoms are mostly empty space.
- The spheres are **inelastic** and the model does not show the **movement** or energy of particles, nor the actual distances between them (gas particles are much further apart than can be drawn).

4.2.2.2 State Symbols

> In chemical equations, the three states of matter are shown as **(s)** solid, **(l)** liquid and **(g)** gas, with **(aq)** for **aqueous solutions** (substances dissolved in water).

Example: $CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

> **Exam Tip:** Water is **(l)** not (aq) — (aq) means "dissolved in water", and water cannot be dissolved in itself. A precipitate formed in a reaction is **(s)**. Gases given off are **(g)**.

4.2.2.3 Properties of Ionic Compounds

Property	Explanation
High melting and boiling points	There are **strong electrostatic forces of attraction** between oppositely charged ions in the giant lattice, acting in all directions. A **large amount of energy** is needed to overcome these many strong bonds. (MgO, with 2+ and

	2– ions, has a higher melting point (2852 °C) than NaCl (801 °C) because the attraction between more highly charged ions is stronger.)
Do not conduct electricity when solid	The ions are held in **fixed positions** in the lattice and **cannot move**.
Conduct when melted or dissolved in water	The ions are **free to move** and so can **carry charge** through the liquid or solution.
Often soluble in water	Water molecules can attract and surround the ions.
Brittle	A force shifts layers so that like charges line up and repel, and the crystal shatters.

Exam Tip: Ionic compounds conduct when molten or in solution because the **ions** move — *not* electrons. Saying "electrons move" for an ionic compound is a classic error that loses the mark. Equally, saying "ionic bonds are broken when NaCl melts" is acceptable, but saying "intermolecular forces are broken" is wrong — there are no molecules in an ionic compound.

4.2.2.4 Properties of Small Molecules

Substances that consist of **small molecules** are usually **gases or liquids** that have relatively **low melting points and boiling points**. These substances have only **weak forces between the molecules** (intermolecular forces). It is these intermolecular forces that are overcome, **not the covalent bonds**, when the substance melts or boils.

- The intermolecular forces **increase with the size of the molecules**, so larger molecules have higher melting and boiling points. This explains the trend in the halogens (F_2 gas → I_2 solid) and

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in alkanes in crude oil.

- Substances that consist of small molecules **do not conduct electricity**, because the molecules do not have an overall electric charge and there are no free electrons or ions.

Substance	Formula	Relative formula mass	Boiling point (°C)	State at 20 °C
Hydrogen	H_2	2	−253	Gas
Methane	CH_4	16	−162	Gas
Chlorine	Cl_2	71	−34	Gas
Bromine	Br_2	160	59	Liquid
Iodine	I_2	254	184	Solid

Exam Tip: This is the single most-tested misconception in the topic. When a simple molecular substance boils, the **weak intermolecular forces** between molecules are overcome; the **strong covalent bonds** inside each molecule are *not* broken. If you write "the covalent bonds are weak" or "covalent bonds break when water boils", you will lose the marks. Water vapour is still H_2O molecules.

4.2.2.5 Polymers

Polymers have **very large molecules**. The atoms in the polymer molecules are linked to other atoms by **strong covalent bonds**. The **intermolecular forces between polymer molecules are relatively strong**, and so these substances are **solids at room temperature**.

Polymers are made by joining together thousands of small molecules called **monomers**. For example, poly(ethene) is made from ethene (C_2H_4) monomers. A polymer can be represented by its **repeating unit** in brackets with "n" after it, showing it repeats many times.

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```
Repeating unit of poly(ethene):

        (  H   H  )
        |  |   |  |
      —|—C——C—|—
        |  |   |  |
        (  H   H  )n

Molecular formula of the repeating unit: C₂H₄
The bonds crossing the brackets show the chain continues on both sides.
```

You should be able to recognise polymers from diagrams showing their bonding and structure. Because the molecules are long, tangled chains, the intermolecular forces add up along their length and are much stronger than those between small molecules — this is why plastics are solid, while ethene itself is a gas.

4.2.2.6 Giant Covalent Structures

> Substances that consist of **giant covalent structures** are **solids with very high melting points**. All of the atoms in these structures are linked to other atoms by **strong covalent bonds**. These bonds must be overcome to melt or boil these substances. **Diamond** and **graphite** (forms of carbon) and **silicon dioxide** (silica) are examples of giant covalent structures.

Silicon dioxide (SiO_2) is the main compound in sand. Each silicon atom is covalently bonded to four oxygen atoms, and each oxygen atom is bonded to two silicon atoms, in a giant 3D lattice. Like diamond, it is hard, has a very high melting point (about 1700 °C) and does not conduct electricity.

4.2.2.7 Properties of Metals and Alloys

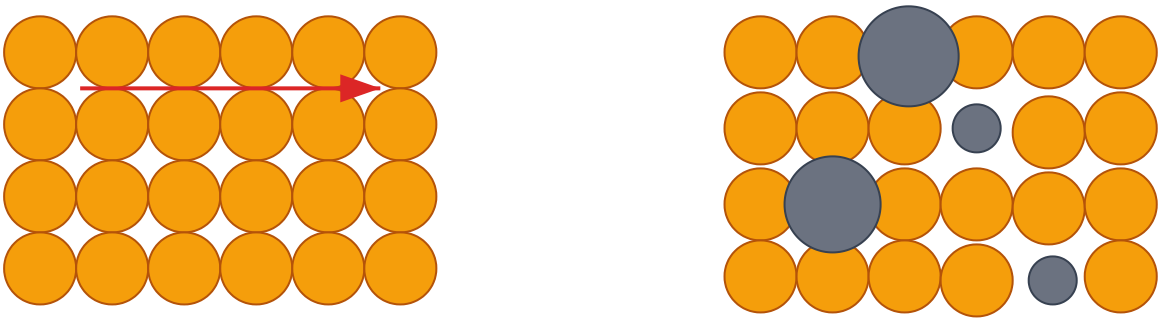
Metals have **giant structures** of atoms with **strong metallic bonding**. This means that most metals have **high melting and boiling points** — a lot of energy is needed to overcome the strong attraction between the positive ions and the delocalised electrons.

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In pure metals, atoms are arranged in **layers**, which allows metals to be **bent and shaped** (malleable) — the layers can **slide over each other** while the delocalised electrons keep holding the structure together. Pure metals are often **too soft** for many uses, so they are mixed with other metals to make **alloys**, which are harder.

An **alloy** is a mixture of a metal with at least one other element (usually another metal, or carbon in steel). The **different sizes of atoms** in an alloy **distort the layers** in the structure, making it **more difficult for them to slide over each other**, so alloys are **harder** than pure metals.

Pure metal vs alloy



Pure metal
regular layers slide easily → soft

Alloy
different-sized atoms distort layers → harder

Common alloys include **steel** (iron + carbon, and sometimes other metals), **brass** (copper + zinc), **bronze** (copper + tin), and **gold alloys** used for jewellery (pure gold is 24 carat; 18 carat gold is 75% gold). Alloys are covered further in topic 4.10 Using Resources.

4.2.2.8 Metals as Conductors

Metals are **good conductors of electricity** because the **delocalised electrons** in the metal carry electrical charge through the metal. Metals are **good conductors of thermal energy** because energy is transferred by the delocalised electrons.

Exam Tip: When explaining why metals conduct, use the word **delocalised** and say what the electrons do: "the delocalised electrons are free to move through the structure and carry

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charge". Simply saying "metals have free electrons" may score only 1 of 2 marks. For thermal conduction, say "energy is transferred by the delocalised electrons".

4.2.3 Structure and Bonding of Carbon

Carbon can form several different structures (called **allotropes**) because each carbon atom can form up to four covalent bonds. The properties of each form depend on how the atoms are bonded together.

4.2.3.1 Diamond

In **diamond**, each carbon atom forms **four covalent bonds** with other carbon atoms in a giant covalent structure, so diamond is **very hard**, has a **very high melting point** and **does not conduct electricity**.

- **Very hard** — every atom is held by four strong covalent bonds in a rigid 3D tetrahedral network. Used in cutting tools and drill bits.
- **Very high melting point** (above 3500 °C) — a huge amount of energy is needed to break the many strong covalent bonds.
- **Does not conduct electricity** — all four outer electrons of each carbon are used in bonding; there are **no delocalised electrons** or ions.

4.2.3.2 Graphite

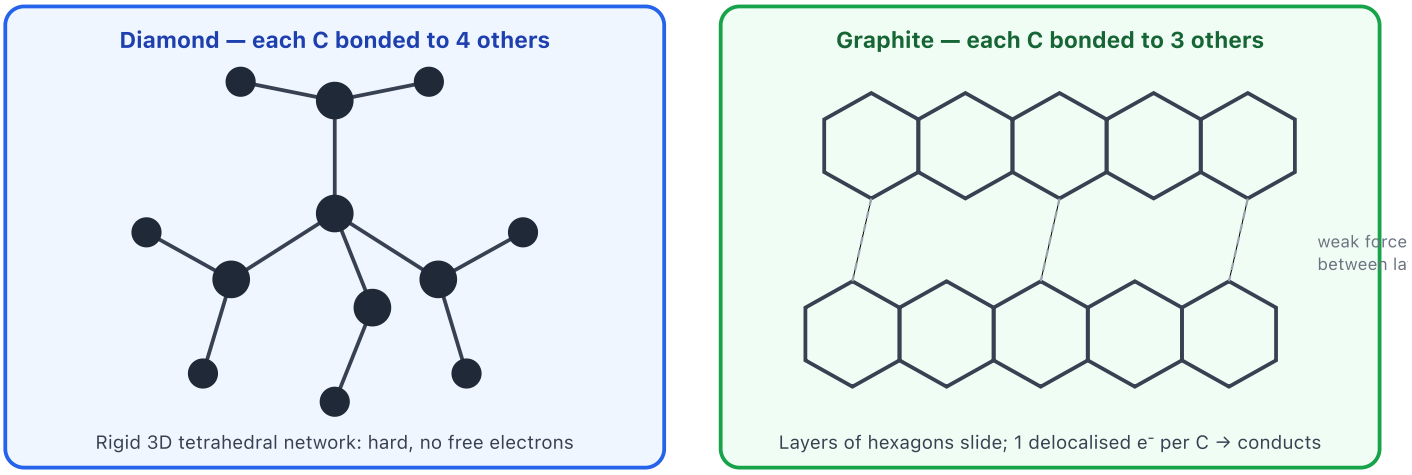
In **graphite**, each carbon atom forms **three covalent bonds** with three other carbon atoms, forming **layers of hexagonal rings** which have **no covalent bonds between the layers**. In graphite, **one electron from each carbon atom is delocalised**.

- **Soft and slippery** — there are only weak intermolecular forces between the layers, so the **layers can slide over each other**. Used in pencils and as a lubricant.
- **High melting point** — the strong covalent bonds within the layers must be broken to melt it.

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- **Conducts electricity and thermal energy** — the delocalised electrons can move along the layers. Used for electrodes. You should be able to explain the properties of graphite in terms of its structure and bonding, and know that graphite is similar to metals in that it has delocalised electrons.

Diamond and graphite: two giant covalent forms of carbon



Comparison of Diamond and Graphite

Feature	Diamond	Graphite
Bonds per carbon atom	4	3
Structure	Rigid 3D tetrahedral lattice	Flat layers of hexagonal rings
Hardness	Very hard	Soft, slippery
Melting point	Very high	Very high
Electrical conductivity	Does not conduct (no delocalised electrons)	Conducts (one delocalised electron per atom)
Uses	Cutting tools, jewellery	Electrodes, pencils, lubricants

4.2.3.3 Graphene and Fullerenes

Graphene is a **single layer of graphite** — one atom thick. It has properties that make it useful in **electronics and composites**. You should be able to explain the properties of graphene in terms of its structure and bonding.

- **Very strong** — held by strong covalent bonds in the layer (about 100 times stronger than steel of the same thickness).
- **Conducts electricity very well** — delocalised electrons move freely across the layer.
- **Transparent and extremely light** — being only one atom thick.

Fullerenes are molecules of carbon atoms with **hollow shapes**. The structure of fullerenes is based on **hexagonal rings of carbon atoms** but they may also contain rings with **five or seven carbon atoms**. The first fullerene to be discovered was **Buckminsterfullerene (C_{60})**, which has a **spherical** shape.

Carbon nanotubes are **cylindrical fullerenes** with very high **length to diameter ratios**. Their properties make them useful for **nanotechnology**, **electronics** and **materials**:

- high **tensile strength** (resist being pulled apart) — used to reinforce materials such as tennis rackets and bicycle frames;
- high **electrical and thermal conductivity** — used in electronics.

Fullerene	Shape	Key properties	Uses
Buckminsterfullerene, C_{60}	Hollow sphere (like a football: 20 hexagons + 12 pentagons)	Simple molecular — weak intermolecular forces, so relatively low melting point and slippery	Lubricants; delivering drugs into the body (molecules can cage other molecules)
Carbon nanotubes	Hollow cylinders	High tensile strength; conduct electricity and heat; huge surface area	Reinforcing composites, electronics, catalysts (large surface area)

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Graphene	Single flat layer	Strong, conductive, transparent, light	Electronics, touch screens, composites

Exam Tip: Fullerenes are **molecules** (C_{60} has a fixed formula), unlike diamond and graphite which are giant structures. So C_{60} has a much **lower** melting point than diamond — when it melts, only the weak intermolecular forces between C_{60} molecules are overcome. Questions often give you data about an unfamiliar form of carbon and ask you to explain its properties — always link each property to the **bonding** (number of covalent bonds per atom, delocalised electrons, forces between layers or molecules).

Summary of Structure Types

Structure	Examples	Melting / boiling point	Conducts electricity?	What is overcome on melting
Giant ionic	NaCl, MgO	High	Only when molten or in solution (ions move)	Strong electrostatic forces between ions
Simple molecular	H_2O, CO_2, Cl_2, CH_4	Low	No	Weak intermolecular forces
Polymer	Poly(ethene)	Moderate (solid at room temp.)	No	Intermolecular forces between chains
Giant covalent	Diamond, graphite, SiO_2	Very high	No (except graphite)	Strong covalent bonds
Giant metallic	Cu, Fe, alloys	Generally high	Yes, solid and liquid (delocalised electrons)	Strong metallic bonds

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4.2.4 Bulk and Surface Properties of Matter including Nanoparticles (Triple Only)

4.2.4.1 Sizes of Particles and their Properties

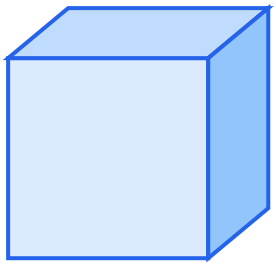
> **Nanoscience** refers to structures that are **1–100 nm** in size, of the order of a few hundred atoms. **Nanoparticles** are smaller than fine particles ($PM_{2.5}$), which have diameters between 100 and 2500 nm (1×10^{-7} m and 2.5×10^{-6} m). **Coarse particles** (PM_{10}) have diameters between 1×10^{-5} m and 2.5×10^{-6} m. Coarse particles are often referred to as **dust**.

Particle type	Diameter (nm)	Diameter (m)
Atoms and small molecules	~0.1–1	$\sim 10^{-10}$–10^{-9}
Nanoparticles	1–100	1×10^{-9} – 1×10^{-7}
Fine particles ($PM_{2.5}$)	100–2500	1×10^{-7} – 2.5×10^{-6}
Coarse particles (PM_{10}, dust)	2500–10 000	2.5×10^{-6} – 1×10^{-5}

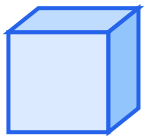
As the side of a cube **decreases by a factor of 10**, the **surface area to volume ratio increases by a factor of 10**. Nanoparticles may have properties **different from those for the same materials in bulk** because of their **high surface area to volume ratio** — a much larger proportion of the atoms are at the surface. It may also mean that **smaller quantities are needed** to be effective than for materials with normal particle sizes.

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Smaller cubes have a larger surface area to volume ratio



Side = 10 nm
SA = 600 nm², V = 1000 nm³
SA : V = 0.6 : 1



Side = 5 nm
SA = 150 nm², V = 125 nm³
SA : V = 1.2 : 1



Side = 1 nm
SA = 6 nm², V = 1 nm³
SA : V = 6 : 1

Worked Example 2 — Surface Area to Volume Ratio

Question: A cube-shaped nanoparticle has sides of 20 nm.
Calculate its surface area to volume ratio.

Surface area = 6 × (side)² = 6 × 20² = 6 × 400 = 2400 nm²
Volume = (side)³ = 20³ = 8000 nm³
SA : V = 2400 ÷ 8000 = 0.3

Answer: SA : V = 0.3 : 1 (or 0.3 nm⁻¹)

If the side were 2 nm (10 times smaller):
SA = 6 × 4 = 24 nm², V = 8 nm³, SA : V = 3 : 1
→ ratio has increased by a factor of 10 ✓

Worked Example 3 — Comparing Particle Sizes

Question: A nanoparticle has a diameter of 25 nm. A fine particle of soot has a diameter of 2.5×10^{-6} m. How many times larger is the soot particle's diameter?

Convert: 25 nm = 25×10^{-9} m = 2.5×10^{-8} m
Ratio: $(2.5 \times 10^{-6}) \div (2.5 \times 10^{-8}) = 1 \times 10^{2} = 100$

Answer: The soot particle is 100 times larger in diameter.

4.2.4.2 Uses of Nanoparticles

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Nanoparticles have many applications in **medicine**, **electronics**, **cosmetics and sun creams**, as **deodorants**, and as **catalysts**. New applications for nanoparticulate materials are an important area of research.

Application	Example	Property used
Sun creams	Titanium dioxide and zinc oxide nanoparticles	Better UV protection and better skin coverage; transparent (no white residue).
Medicine	Fullerene "cages" or nanoparticles for targeted drug delivery	Tiny size allows them to be absorbed by cells; can carry drug molecules to specific sites.
Antibacterial products / deodorants	Silver nanoparticles in socks, wound dressings, surgical masks, fridges	Silver is antibacterial; high surface area makes it very effective in small amounts.
Catalysts	Nanoparticulate platinum, gold, nanotubes as catalyst supports	Huge surface area to volume ratio — more active sites per gram; less expensive metal needed.
Electronics	Carbon nanotubes and graphene in tiny circuits	Conduct electricity; extremely small.
Cosmetics	Moisturisers and make-up	Small particles penetrate skin more easily and do not leave residue.
Materials	Nanotube-reinforced sports equipment	High strength-to-weight ratio.

Risks of Nanoparticles

You should be able to evaluate the **benefits and disadvantages** of nanoparticles:

- Their effects on health are **not yet fully understood** — they are a relatively new technology with little long-term data.

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- Because they are so small, they may be **absorbed through the skin** or **breathed into the lungs** and enter cells, where they might cause damage (possibly being toxic or catalysing harmful reactions).
- Nanoparticles washed into the **environment** (e.g. silver from clothing) might harm useful bacteria in water treatment or ecosystems.
- Many argue that products containing nanoparticles should be **tested more thoroughly** and **clearly labelled**.

> **Exam Tip:** For 6-mark "evaluate the use of nanoparticles in sun cream" questions, give a **balanced** answer: benefits (better UV protection, better coverage, no white marks, so may reduce skin cancer risk) and risks (may be absorbed into cells, long-term effects unknown, may damage the environment when washed off), then a **justified conclusion**. Always explain the benefit using the **high surface area to volume ratio** where relevant.

Summary Checklist for 4.2 Bonding, Structure and the Properties of Matter

- Identify ionic, covalent and metallic bonding from the elements involved.
- Explain ionic bonding as electron transfer forming ions with noble gas structures; draw dot and cross diagrams for ionic compounds.
- Work out the charges of ions in Groups 1, 2, 6 and 7, and deduce formulae of ionic compounds (including compound ions).
- Describe the structure of a giant ionic lattice and the limitations of different models; deduce empirical formulae from diagrams.
- Explain covalent bonding as shared pairs of electrons; draw dot and cross diagrams for H_2, Cl_2, O_2, N_2, HCl, H_2O, NH_3 and CH_4.
- Describe metallic bonding: positive ions and delocalised electrons.
- Use the particle model to explain changes of state; predict states from melting and boiling point data. (Higher: limitations of the particle model.)
- Use state symbols (s), (l), (g), (aq) correctly.

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- Explain the properties of ionic compounds, small molecules, polymers, giant covalent structures and metals in terms of structure and bonding.
- Explain why alloys are harder than pure metals, and why metals conduct electricity and heat.
- Explain the properties of diamond, graphite, graphene, fullerenes (C_{60}) and carbon nanotubes.
- (Triple) Compare the sizes of nanoparticles, fine and coarse particles; calculate surface area to volume ratios.
- (Triple) Describe uses of nanoparticles and evaluate their benefits and risks.

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